FRANK DUMOLUHLE NCUBE

Salisbury, NC | frankdumoluhle24@gmail.com | (743) 542-0844
linkedin.com/in/frank-ncube-417a52338 | github.com/Knarf24 | frank-ncube-portfolio.vercel.app

EDUCATION

Livingstone College | B.S. Computer Information Sciences **Expected December 2029**
Salisbury, NC | GPA: 4.0/4.0 | International Presidential Scholar
Relevant coursework: Programming Fundamentals, Computer Information Systems, Networks & Telecommunications, College Mathematics

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS
Web & data: Next.js, React, Node.js, Express, PostgreSQL, Supabase, REST APIs, Vercel, Tailwind CSS
AI & engineering: LLM API integration, retrieval (TF-IDF), prompt engineering, Git/GitHub, Vitest, AWS fundamentals

SELECTED TECHNICAL PROJECTS

ShopAccess AI | AI shopping assistant with permission-based checkout **2026**
PayPal AI Hackathon | Next.js, React, TypeScript, Zod, Vitest, Chrome Manifest V3, PayPal Sandbox

- Built a bilingual (English/Spanish) laptop-shopping prototype that turns natural-language needs into shopper-confirmed requirements and server-validated recommendations from a demo catalog.
- Separated advisory AI output from deterministic server checks: AI-hardware compatibility is shown as “not verified” without evidence, and checkout requires a revocable, item-specific purchase permission with quantity, total, and expiry limits.
- Added a paired Chrome extension side panel (cannot confirm or pay) and an optional PayPal Sandbox flow; default checkout is simulated. 2,568 automated tests pass locally.

EtenMall | Multi-tenant e-commerce platform (in development) **2026–Present**
Next.js, TypeScript, Supabase, PostgreSQL, Vercel

- Architected a tenant-isolated retail platform (storefronts, branch inventory, product variants, staff roles, guest checkout) using layered services/repositories, PostgreSQL row-level security, and tested privilege boundaries across 83 migrations.
- Reproduced and fixed a PostgreSQL checkout deadlock via deterministic lock ordering; shipped staged, flag-gated releases with audit logging and test-mode Paynow payments (not live). 4,412 unit tests pass locally.

Triage360 | AI support triage system **2026**
React, Express, PostgreSQL, TypeScript, Python, Claude API

- Built ticket classification, document retrieval, and grounded AI responses with human escalation for six risk categories.
- Added ticket history and statistics; built a separate Python prototype exploring TF-IDF and semantic retrieval.

Horizon Desk | AI banking relationship prototype **2026**
SteelHacks XIII (team of three) | Next.js, TypeScript, Claude 3.5 Sonnet, ElevenLabs, Vitest

- Co-developed the frontend and supporting backend for a synthetic-data prototype that applies deterministic eligibility checks before AI-assisted outreach and routes credit opportunities to human review.

EXPERIENCE

Livingstone College Academic Support | Math & science tutor **2026–Present**

- Tutor 10+ students in individual and small-group sessions, adapting explanations and practice to learning needs.

Frank Wi-Fi Hotspot Business | Founder & operator **2024–Present**

- Operate a Starlink-powered voucher internet service in Gweru, Zimbabwe; coordinate approximately 30 people across sales, installation, marketing, and maintenance, and troubleshoot network issues remotely.

CERTIFICATIONS, TRAINING & LEADERSHIP

AWS Academy Graduate: Cloud Foundations; Generative AI Foundations (2026)
Datafrik Advanced AI & MLOps Bootcamp — In progress (11 weeks): model evaluation, neural networks, NLP, API deployment, Docker, CI/CD, monitoring
Leadership: HBCUniverse Campus Ambassador; HBCU Heroes participant